Enhanced Crop Mapping Using Polarimetric SAR Features and Time Series Deep Learning: A Case Study in Bei'an, China

Niantang Liu, Qunshan Zhao, Richard Williams, Si-Bo Duan, *Senior Member, IEEE*, Xiangyang Liu, *Member, IEEE*, and Brian Barrett

***Abstract*— Large-scale crop mapping is essential for decision-makers to evaluate agricultural resource usage and estimate crop yields. Considering the utility of annual crop inventory (CI) statistics for monitoring crop growth, generalizing near real-time crop classification over large areas becomes necessary. Accurate crop-type identification using remote sensing data remains challenging due to the variability in crop growth patterns across time and space, the presence of crops with similar phenological stages, and the scarcity of labeled data. This study develops deep learning-based approaches to map agricultural regions at the county level using multitemporal Sentinel-1 synthetic aperture radar (SAR) data, specifically evaluating the contribution of SAR-derived input predictors for discriminating both majority and minority crops in Bei'an, Northeast China. The proposed model architecture amalgamates 1-D convolutional layers (Conv1D) with attention-based long short-term memory (LSTM) to characterize the crop types exhibiting phenological similarities using a range of SAR-derived input predictors. The results are compared with alternative multitemporal deep learning frameworks, including standalone Conv1D and Transformer models, as well as the machine learning algorithm random forest (RF), which serves as the baseline for comparison. The designed architecture (Conv1D-LSTM) achieved the highest $F1$ scores (maize: 87%, soybean: 86%, and other crops: 85%) when applied to an inherently imbalanced dataset, using m-chi decomposition features as input predictors. The results provide superior performance in terms of effectiveness and efficiency compared to other selected models. The monthly in-season crop classification underscores the importance of temporal dependencies and the availability of multitemporal observations for learning dynamic growth patterns over large areas. Furthermore, the interpretation of model learning processes and outcomes is explained by visualizing weight distributions and hidden features. This study offers a comprehensive evaluation of essential SAR features in multitemporal satellite data for accurate crop mapping, utilizing advanced deep learning techniques. This work is available at https://github.com/Niantangliu/Deep-learning-crop-mapping.**

***Index Terms*— Crop mapping, deep learning, multitemporal satellite data, polarimetric synthetic aperture radar (PolSAR).**

I. Introduction

ACCURATE crop mapping is essential for dynamically monitoring agricultural productivity, assessing food availability, and supporting decision-making at regional and national levels [1], [2], [3], [4]. For instance, early season crop classification contributes to numerous agricultural applications, such as cropland management, supply chain frameworks, crop insurance, and area-based subsidies supported by governments [5]. Accurate and near real-time crop classification is greatly needed for understanding changes in growth patterns of specific crops and assessing their socioeconomic impacts.

Modern satellite remote sensing technology enables the detection, mapping, and monitoring of agronomic information over various spatial and temporal scales and can help optimize agricultural practices, reducing the environmental impact of food production and minimizing waste, and ultimately contributing to more sustainable food systems [6], [7]. Satellite remote sensing offers opportunities to obtain multidimensional data at high spatial and temporal resolutions over large regions. Satellite-based workflows have been widely employed to enhance crop mapping accuracies and produce crop maps at various scales [8]. For example, cropland data layer (CDL) and crop inventory (CI) are two national-scale, medium-resolution crop map products that are updated annually. The 2022 CDL, a geo-referenced raster layer of the United States Department of Agriculture (USDA), offers geospatial information pertaining to crop types, land cover, and land use at a 30-m resolution across USA [9]. In contrast, CI is developed by Agriculture and Agri-Food Canada (AAFC) and provides crop-type information for Canada at a 30-m resolution [10]. Generating both products involves the integration of multisource satellite imagery and comprehensive ground-truth data collected via field surveys.

Although the multispectral and photosynthetic properties of vegetation can be measured by optical satellite platforms [11], frequent cloud cover can limit the quality of optical acquisitions, significantly impacting the performance of effective

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differentiation of crops [12], [13]. Methods to reconstruct occluded information, such as linear interpolation [14], and probabilistic models such as K-nearest neighbor (KNN) imputation and decision trees [15], are not only computationally demanding but also encounter challenges in achieving accurate imputation, particularly for large datasets with significant missing or contaminated data [16].

To address these limitations, synthetic aperture radar (SAR) sensors, active instruments that can operate independently of illumination and weather conditions, have been widely used to retrieve information on the Earth's surface. Previous studies have demonstrated that time series SAR features positively impact land cover classification performance (e.g., [17], [18]). However, their applications in agricultural domains remain challenging compared to the utilization of optical data [19], [20] due to the complex scattering mechanisms of SAR signals in agricultural landscapes. The intricate interactions between SAR signals, crop canopy structure, soil moisture, and surface roughness necessitate advanced processing techniques to extract meaningful crop information and utilize the potential of SAR-based features for accurate crop-type identification.

The Sentinel-1 satellite constellation, with a short revisit frequency of six days when both satellites are operational, offers open-access SAR data at C-band with global coverage, ensuring consistent monitoring and mapping applications in agriculture [21], [22], [23], [24], [25]. However, most studies have focused primarily on radar amplitude information as input predictors. Sentinel-1 can also provide phase information characterized by off-diagonal elements in the scattering matrix, which can improve crop mapping accuracy by investigating the scattering properties of different land cover types [26], [27]. In addition, polarimetric SAR (PolSAR) is an advanced radar remote sensing technique that measures and interprets the polarization state of backscattered signals, thereby enabling a detailed understanding of the scattering mechanism and enhancing the identification and characterization of surface features, such as crops. Polarimetric decomposition algorithms have been developed to extract physical information from the target surface, which can break down the random scattering mechanisms into several parameters that can be associated with the target's physical characteristics. The polarimetric parameters are also sensitive to crop phenological changes [28]. However, many existing decomposition methods are only applicable to quad-polarization (quad-pol) sensors [29], [30], [31], [32], which limits their use with dual-polarized platforms like Sentinel-1. Furthermore, quad-polarimetric observations often have reduced swath coverage and limited availability, hindering their application across large areas with high temporal frequency. While dual-polarization (dual-pol) data have been investigated as an alternative solution for large-scale crop monitoring by several studies [27], [33], [34], [35], [36], there remains a pressing need for more refined algorithms that can accurately capture variability both within fields and across landscapes.

Recently, deep learning techniques have shown their efficiency and reliability in handling remote sensing data for a range of applications such as land cover classification, crop-type classification, plant disease detection, and crop yield prediction [30], [37], [38], [39]. These cutting-edge methods employ end-to-end neural network architectures capable of inherently exploring and learning high-dimensional information. Convolutional neural networks (CNNs) are extensively employed to extract multiscale features from remote sensing data [40]. Specifically, 1-D CNNs (1D-CNNs) have been successfully employed for crop mapping by using 1-D convolutional layers (Conv1D) layers as pixel-based feature extractors processing multitemporal remote sensing datasets [41], [42]. In addition, recurrent neural networks (RNNs), long short-term memory (LSTM), and Transformers have also demonstrated success in enhancing crop mapping by extracting dynamic temporal features from longer image sequences [43], [44], [45]. In the case of stacked Conv1D layers, the shallow layers extract local features, while deeper layers comprehensively summarize patterns to a larger extent. LSTM units, on the other hand, are designed to memorize features over long- or short-time sequences. It has been shown that deep learning methods are capable of learning hierarchical features across spatiotemporal dimensions and frequently outperform traditional classifiers that rely on manually defined rules for crop classification tasks [26], [46], [47].

In this context, there is a clear need for studies that investigate the potential of deep learning approaches and understand the interpretability of these networks, specifically those designed to uncover sequential relationships in multitemporal remote sensing data for crop mapping. This is particularly important for regions like Northeast China, where obtaining quantitative information from local croplands has been challenging due to annual crop rotation practices [48]. The Northeast region of China has become a significant agricultural region, playing a key role in both domestic agricultural production and international trade, particularly in the cultivation of economically important crops such as soybeans and maize [49], [50]. As a result, there is a strong demand for accurate annual crop maps to assist local authorities in establishing near real-time crop monitoring systems for early yield assessments. However, reliable crop mapping is difficult due to the complex relationship between the location-specific factors (spatial) and the timing-related factors (temporal) that affect crop growth [27], [30], [51], [52]. Although 1D-CNNs benefit from lower computational complexities [53], their predictive performance depends on the sensitivity of input predictors derived from satellite data and the design of model architectures [54], [55]. Crop mapping challenges are further magnified by the limited availability of accurate crop-type labels, resulting in imbalanced dataset distributions and reduced crop map accuracies on a large scale, despite the improvements in spatial and temporal resolutions provided by contemporary Earth-observing satellites [56]. Moreover, labor costs, expertise requirements, and the accessibility of multitemporal remote sensing data continue to pose constraints for large-scale crop monitoring initiatives [57], [58].

The aim of this study is to develop an approach for multitemporal crop mapping in Bei'an, Northeast China, using PolSAR-derived data and deep learning. The research addresses the following research questions.

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1) How does the developed Conv1D-LSTM architecture perform in comparison to existing models, such as Conv1D [41], Conv1D-random forest (RF) [54], Transformer [45], and the universal baseline model RF, with respect to crop mapping performance and handling imbalanced class distribution?
2) Which SAR features derived from Sentinel-1 datasets have the greatest impact on crop mapping in Bei'an?
3) How do phenological factors influence the effectiveness of in-season crop mapping using SAR data?
4) How is the interpretability of the proposed deep learning approach demonstrated in this study?

Specifically, the study develops a joint ensemble learning architecture (Conv1D-LSTM) by combining the attention-based LSTM and Conv1D layers to reduce the recognition error rate resulting from the imbalanced dataset. The developed architecture is compared to existing state-of-the-art deep learning models and the baseline model for differentiating local major crops such as soybean and maize, which exhibit similar phenological stages. In addition, we explore various SAR features, including the backscatter coefficient, gray-level co-occurrence matrix (GLCM), covariance matrix parameters, radar vegetation index (RVI), dual-pol RVI (DpRVI), and polarimetric features derived by m-chi decomposition and dual-polarization entropy/alpha. Due to the potential for high-dimensional SAR features to introduce redundant information, we explored feature selection techniques, such as the Spearman correlation coefficient and the feature importance algorithm (Boruta), to eliminate less relevant features. We then created multiple scenarios to evaluate the importance of specific SAR features in differentiating crops.

Section II describes the data used in this study. Section III outlines the methodology for feature selection and classification model development. Section IV analyzes the results, identifies optimal SAR features, and interprets the deep learning models. Finally, Section V summarizes the key findings and implications.

II. Material

Bei'an, located in the transitional zone between the Songnen Plain and the Greater Khingan mountains (47°35′N–48°33′N and 126°16′E–127°53′E), is a county-level city in the northeast part of Heilongjiang, China (see Fig. 1, left). Known for its fertile black soil, Bei'an is ideal for growing crops. About 35.4% of the area is forested, and 32.95% is cropland, mainly for spring soybean (61.8%) and maize (29.5%) [59], with minor cultivation of other crops, including rice and wheat. The entire croplands exhibit complex cropping patterns that pose potential challenges for satellite-based crop mapping, particularly when relying on medium- or low-spatial-resolution imagery.

The proportion of reference samples for each class and the field size distribution are illustrated in Fig. 2. The field polygons were surveyed in 2017 by Chinese Academy of Agricultural Sciences (CAAS). Narrow, strip-like field parcels were classified using high-resolution imagery, such as the 5-m RapidEye data, while larger crop parcels were manually labeled using 10-m resolution Sentinel-2 imagery.

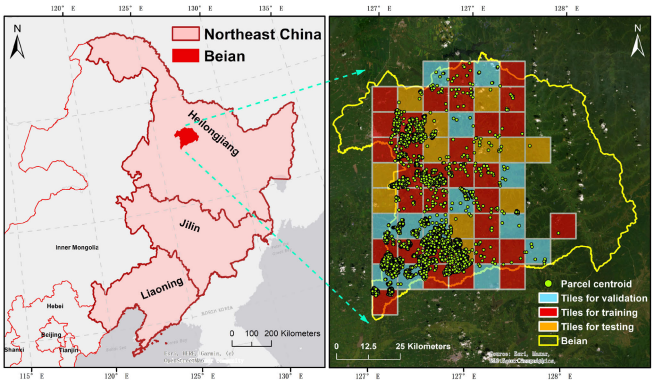


Fig. 1. Study area of Bei'an, (left) Northeast China, and (right, centroids used for visualization clarity) data partitioning based on 10-km tiles with the spatial distribution of parcel centroids.

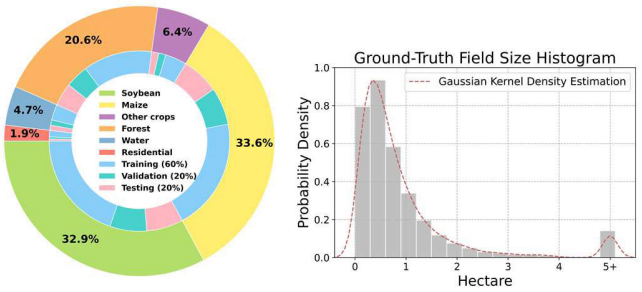


Fig. 2. (Left) Proportion of imbalanced samples for each class. The training, validation, and testing datasets are mutually exclusive. Pixels from each class were extracted from these sets in a ratio of 60:20:20. (Right) Distribution of crop field size in the Bei'an ground-truth dataset. Parcels larger than 4 ha are aggregated in the last bin of the histogram. The average parcel size is 1.39 ha.

This process resulted in a comprehensive dataset comprising 3 979 417 labeled pixels. The red, vegetation red edge, and SWIR bands from Sentinel-2, along with the red, red edge, and NIR bands from RapidEye, enabled significant differentiation between maize and soybean [57]. This dataset was then allocated for training (60%), validation (20%), and testing (20%) using 10-km grids (see Fig. 1, right), following the data partitioning approach [41]. A random 10% of each dataset subset (equivalent to 397 942 pixels) was selected for dual purposes: conducting feature selection to eliminate irrelevant features and training models with selected features in a weakly supervised manner. Subsequently, comparing the model performance leads to the most effective SAR-related predictors for crop mapping. After identifying key predictors, the remaining ground-truth labels were reintegrated into the 10% dataset to retrain the models and assess accuracy improvements in crop mapping.

According to the local crop sowing scheme, maize is typically sown in late April, whereas soybeans are normally sown in early May. Data collection focused on the period from early May to late September (May 6 and September 20). In total, 22 Sentinel-1B acquisitions were exclusively acquired from 2017 during the growing season, as it was the only Sentinel-1 satellite with available coverage over our study area in the Copernicus database. Fig. 3 illustrates this study's workflow. Fig. 4 shows the monthly distribution of acquisitions, along

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TABLE I
INPUT FEATURES AND BANDS FOR FEATURE SELECTION

Input feature per date	Bands
Backscatter	VV
	VH
	VH/VV
GLCM	VV_mean
	VV_variance
	VV_correlation
	VH_mean
	VH_variance
	VH_correlation
Dual-pol covariance matrix	C_{11}
	C_{22}
	C_{12} real
	C_{12} imagery
Dual-pol Radar vegetation index	DpRVI
Radar vegetation index	RVI
H/α dual-pol decomposition	Alpha
	Entropy
m-chi decomposition	Single-bounce scattering
	Double-bounce scattering
	Volume backscattering

with the corresponding growth stages of soybean and maize in 2017. The phenological stages for maize and soybean are synthesized in monthly intervals according to [56], with the corresponding crops' appearance at each phenological stage defined by Biologische Bundesanstalt, Bundessortenamt und Chemische Industrie (BBCH)-scale codes [60].

The preprocessing of Sentinel-S1B images generally includes radiometric calibration, polarimetric matrix generation, multilooking, polarimetric speckle filtering, and geocoding. Intensity data were converted into backscatter coefficient sigma nought (σ^0) in a logarithmic decibel scale. The cross-ratio of the backscatter is calculated by VH minus VV in terms of the logarithm rules. SAR scattering features and DpRVI were derived from the dual-pol covariance matrix (C_2) using Sentinel-1 Toolbox in SNAP 8.0.0. Based on C_2, this study evaluated decomposition features derived by dual-polarization entropy/alpha decomposition (H/α dual-pol decomposition) and m-chi decomposition. All processed features were georeferenced using the digital elevation model (DEM) from the Shuttle Radar Topography Mission (SRTM 3 arcsecond) and then resampled to a 10-m spatial resolution (see Table I).

III. METHOD

A. SAR-Derived Features

Sentinel-1 single look complex (SLC) products provide dual-pol information in a covariance matrix with two orthogonal components of the scattered signal. Various studies (e.g., [61], [62]) have explored the scattering mechanisms of target features for crop classification using modified H/α dual-pol decomposition tailored for dual-polarized Sentinel-1 data, defined as

$$H = \sum_{i=1}^{2} -P_i \log_2 P_i, \quad \text{wherein } P_i = \frac{\lambda_i}{\sum_{i=1}^{2} \lambda_i} \text{ and } \lambda_1 \geq \lambda_2 \tag{1}$$

$$\alpha = \sum_{i=1}^{2} P_i \alpha_i, \quad \text{wherein } 0 \leq \alpha \leq 90^\circ \tag{2}$$

where λ_1 and λ_2 are eigenvalues of the covariance matrix. The polarimetric scattering entropy H ranging from 0 to 1 is an indicator that measures the degree of scattering randomness. The mean alpha angle α indicates physical scattering characteristics. Based on both parameters, an H/α plane is proposed to represent all random scattering mechanisms and utilized for crop growth analysis [61].

The m-chi decomposition is a compact polarimetric decomposition method developed by Raney et al. [62] to analyze scattering characteristics from compact PolSAR data. This approach facilitates extracting scattering information from the target while reducing the number of transmitted and received channels compared to quad-pol systems. Using the covariance matrix constructed from Sentinel-1 data, the m-chi decomposition generates "pseudo" polarimetric features resembling compact polarimetric parameters. Three parameters for m-chi decomposition can be derived as follows:

$$B = \sqrt{\frac{mS_1(1 - \sin 2\chi)}{2}} \tag{3}$$

$$R = \sqrt{\frac{mS_1(1 + \sin 2\chi)}{2}} \tag{4}$$

$$G = \sqrt{S_1(1 - m)} \tag{5}$$

where B, G, and R correspond to single-bounce, double-bounce, and volume backscattering, respectively; S_1 indicates the stoke parameter for each pixel in the total power over an image field; m is the degree of polarization (DOP); and $\sin 2\chi$ is the degree of circularity. The DpRVI has previously been demonstrated to exhibit a strong correlation with crop biophysical variables and effectively represents crop growth dynamics. To quantitively assess the scattering strength, Mandal et al. [64] introduced a parameter β, which is subsequently used to derive DpRVI

$$\beta = \frac{\lambda_1}{\text{Span}} \tag{6}$$

$$\text{DpRVI} = 1 - \text{DOP} * \beta, \quad \in [0, 1] \tag{7}$$

where λ_1 is the eigenvector obtained through the eigendecomposition of the 2×2 covariance matrix C_2, which are normalized with the total power Span, and DOP indicates the degree of polarization.

The GLCM evaluates the arrangement of grayscale values in an image and is utilized to quantify the variations in texture within images. It calculates spatial features inherent in raster images by determining the probability of occurrence of pixel pairs separated by a predefined distance in a given direction [65]. These features are derived from the properties of pixel co-occurrence statistics, calculated within a specified moving window and following certain directions at interpixel distances. It was found that combining SAR backscatter with statistical group features such as GLCM mean, variance, and correlation led to better crop classification performance [66].

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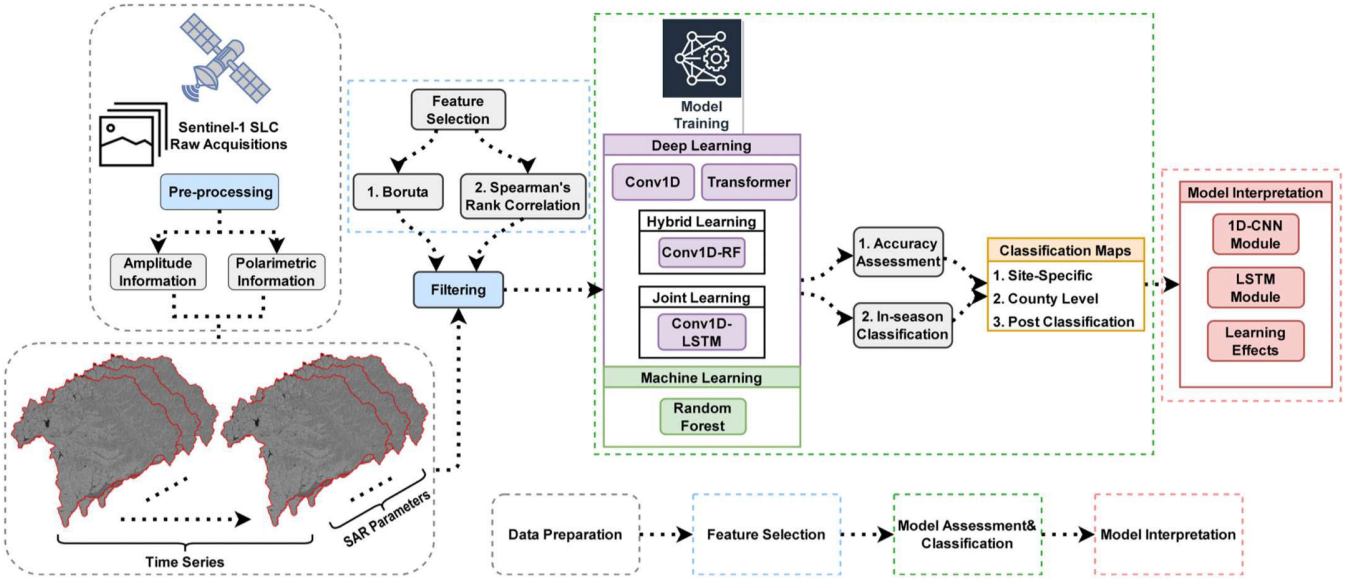


Fig. 3. Methodology flowchart.

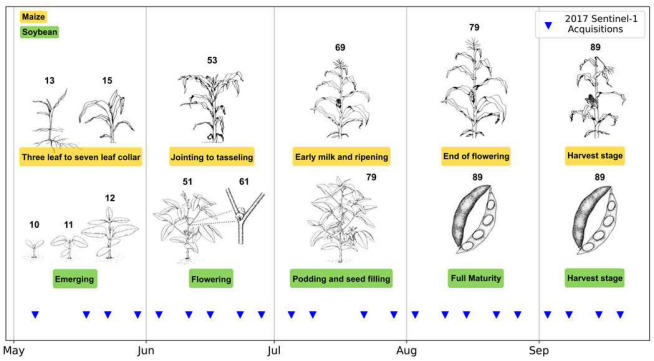


Fig. 4. Sentinel-1 acquisitions, obtained from May 6 to September 20, were analyzed alongside the phenological stages for maize and soybean. The BBCH codes are displayed at the top of the crops' phenological stage.

B. Feature Selection

The practical model building often encounters limitations when working with modern datasets containing multivariate variables. Many of these variables may be irrelevant to the target classification, making it challenging to determine their correlations. For instance, a large PolSAR image tends to contain data redundancy, especially when generating rich indices from multisource remote sensing data [30], [54]. Certain vegetation indices can serve as more effective input features than spectral data for deep learning-based crop classification [41]. As a result, selecting features that strongly correlate with ground objects is important for enhancing classification accuracy. Feature selection typically considers two aspects, including the multicollinearity between input variables and the correlation between the inputs and the targets. Generally, data redundancy can lead to increased computational costs during model training and may negatively impact performance when the optimal feature combination is uncertain. Hence, feature selection techniques that derive minimal and optimal feature sets are essential for achieving the best possible classification results.

First, this study applied the Boruta algorithm, a wrapper method that iteratively compares the importance of each feature in an RF model to the highest importance achieved by randomly permuted copies of those features. It performs a recursive process on multiple feature sets and iteratively eliminates the features less relevant to the label target prediction. This study used an alternative Python-compatible version that leverages Gini impurity to determine feature importance scores, accessed at https://github.com/scikit-learn-contrib/boruta_py. This method evaluates the correlation between inputs and corresponding label targets as the first step.

The second step investigates the relevance between multivariant variables by calculating the Spearman rank correlation coefficient [67]. The Spearman correlation coefficient r_s is given by

$$r_s = \frac{\sum_{i=1}^{N}(a_i - \bar{a})(b_i - \bar{b})}{\sqrt{\sum_{i=1}^{N}(a_i - \bar{a})^2}\sqrt{\sum_{i=1}^{N}(b_i - \bar{b})^2}}, \quad \in [-1, 1] \quad (8)$$

where N is the total number of a feature across the whole temporal dimension and a_i and b_i are distinctive features to be measured. The value range of r_s quantifies the correlations and strength between variables, which can be either negatively or positively correlated.

In this study, the coefficients from multitemporal SAR features previously determined by Boruta are assessed. Based on the feature importance analysis provided in the first step, the number of features can be optimally combined for model training.

C. Classification Approaches

RF, introduced by Breiman [68], has proven to be a widely used machine learning approach for handling high-dimensional remote sensing data in crop mapping. RF is

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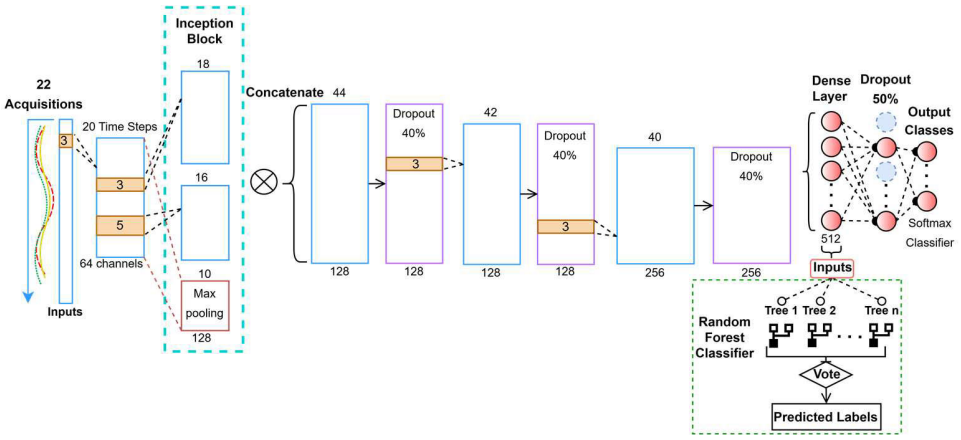


Fig. 5. Architecture of Conv1D and Conv1D-RF. Conv1D processes time series inputs and determines output classes with the softmax classifier. Conv1D-RF extracts inputs from the dense layer and predicts using the RF classifier.

an ensemble classifier that operates multiple unbiased decision trees with a bagging strategy to control the prediction variance of the model performance. However, since RF is not designed to measure time series dependencies, the input features for RF in this study are only divided into a 2-D input shape consisting of the number of pixel-level features and their corresponding bands. To optimize the modeling performance, hyperparameters are tuned to enhance and maintain the model generalization. In this study, the optimal number of trees is set to 200 and the minimal number of samples required to split an internal node is set to 4.

Conv1D is one of the variants of CNN, which employs 1-D filters convolving along the temporal dimension to extract patterns over a time series sequence. Shallow levels of the stacked Conv1D layers concentrate on feature learning at a local scale, while deeper layers summaries learned features holistically, extending to larger extents. However, the final configuration of the CNN layers is initially tested on simple structures and then empirically developed with increased complexity to achieve stable model performance. In this study, two Conv1D-based networks for crop classification performance comparison are applied, as depicted in Fig. 5.

One of the architectures is taken from [41] to extract multitemporal patterns at multiple scales using the inception module. Another architecture Conv1D-RF was derived from [54]. This hybrid Conv1D structure replaces the originally applied softmax classifier with RF in the last fully connected layer, producing final outputs via the unique voting mechanism designed for RF. The Conv1D module in Conv1D-RF needs to be pretrained. It then outputs a feature vector (512×1) from the first fully connected layer, which is subsequently fed into the RF classifier for the final classification.

The LSTM model [69], an extension of the conventional RNN, incorporates the concept of gates to synergistically process time series data with long-term dependencies, retaining temporal features from the input sequence. Temporal features are stored in the LSTM cell at the current time step and passed to the next cell according to weights assigned by each gate, enabling the realization of sequential dependencies. This gate-to-gate delivery mechanism maintains the temporal features with previous and current states to a certain extent in each LSTM cell, allowing for long- and short-term memory formation.

For attention-based LSTM, the attention block refines aggregated hidden features learned by LSTM layers across a long time series sequence for crop mapping by normalizing weights [51], [52].

The Transformer is a recent deep learning model for processing sequential data in natural language processing (NLP) [70]. It has also been used for multitemporal crop classification with optical data in agricultural remote sensing [45], [51], [52]. Fig. 6 illustrates the custom Transformer network used in this study, consisting of positional embedding, encoding blocks, and a multilayer perceptron (MLP) unit. Unlike NLP tasks, processing time series remote sensing data do not require text embedding. Instead, a 1-D position embedding encodes and correlates time steps in the satellite data. These embeddings are linearly transformed for self-attention computation. The self-attention mechanism, utilizing queries, keys, and values, captures various relationships within the input data. Multiple attention heads allow the model to consider different aspects of the data simultaneously. However, self-attention can increase memory requirements for longer sequences. To optimize the performance, residual skip connections and normalization layers are applied around the multihead attention layer. The output is then processed by a feedforward module with linear layers and nonlinear activation functions, maintaining position-wise parallelism and enabling efficient long-sequence processing. The encoder block outputs are globally averaged over time and fed into dense layers for final prediction. Each encoder block has a head size of 256 and eight attention heads, with a total of two encoder blocks followed by an MLP unit.

In this study, we proposed and developed a joint ensemble learning strategy for improved crop classification performance, inspired by Zhou [71] and its successful application in human activity recognition [72]. This method combines Conv1D and LSTM as parallel base learners for temporal feature extraction using customized internal components (see Fig. 7). This joint combination facilitates a mutually complementary effect by

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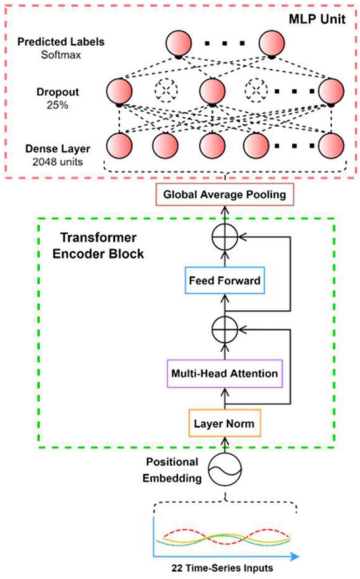


Fig. 6. Transformer-based network.

fusing multisource features given by each module, thereby constructing a robust ensemble feature extractor for sequential data and mitigating model bias toward the majority class. First, we adapt the inception module design for processing multiscale features [41], [54] within the 1D-CNN module. We implement three Conv1D layers, each with kernel sizes of 3, 3, and 5, respectively, with respect to incremental channel depths 64, 128, and 256 for feature learning at different scales, and use the skip connection technique at each level to alleviate the vanishing gradient problem. Batch normalization (BN) is applied at the end of each Conv1D layer to accelerate model convergence. The output features from each level are fed into a fully connected layer and concatenated together.

Second, the LSTM module employs an attention-based mechanism with bidirectional LSTM cells for comprehensive temporal information capture. The attention block aggregates sequential outputs from bidirectional LSTM, producing normalized attention weights to improve classification performance. A fully connected layer adjusts the outputs to 512 units, equal to the last output from the 1D-CNN module.

Finally, the learned features from each module are concatenated and passed through a shared fully connected layer, creating a strong ensemble feature extractor. A dropout layer prevents overfitting and the final output layer generates class probabilities. The joint learning optimization, Conv1D-LSTM, maximizes parallel processing utilization for enhanced time series crop-type prediction.

D. Model Implementation

Three performance indicators were chosen to evaluate crop mapping performance: overall accuracy (OA), Cohen's kappa coefficient (Kappa), and $F1$ score. OA assesses the model's performance for all classes overall, as it represents the ratio of correctly classified samples to the total number of samples. Kappa adjusts classification scores by accounting for the probability of random chance prediction based on observed class frequencies in the dataset. The $F1$ score, which is the harmonic mean of recall and precision, measures classification performance for individual crop types. The deep learning model configurations are modulated with reference to the Adam optimizer and the cross-entropy loss function. The optimizer's learning rate is set at 0.001. The rectified linear unit (ReLU) activation function is employed in all fully connected layers to enhance output nonlinearity and manage systematic errors (bias). In addition, early stopping serves as a regularization strategy, halting model training when performance stabilizes during the iterative training process.

All deep learning models shared these configurations during the training stage. The modeling environment is implemented using Python 3.7.15 and Tensorflow 2.9.2, along with the Keras library (version 2.12) on a Windows 10 system equipped with two NVIDIA Quadro P4000 graphic devices (8-GB RAM per GPU) and two Intel[1] Xeon[1] Silver 4114 CPU processors (2.20/2.19 GHz).

IV. Results and Discussion

A. Temporal Profiles of Sentinel-1 SAR Features

The temporal profiles of Sentinel-1-derived SAR features for each class, as displayed in Fig. 8, show the degree of separation between the crops, which appears challenging, as evidenced by the overlapping areas in the buffer zones across the full growing season for all SAR features. The similarity in the time series profiles of the crops highlights the difficulty in differentiating between crop types with similar phenological cycles based on SAR-derived feature values. Although the values generally follow the growth trends toward their peaks around August, the separability between crops is not distinctly evident in specific SAR-based features. Some features, such as "VH GLCM Mean," "VH GLCM Variance," "Entropy," and "Alpha," exhibit fewer overlaps between August and September, during which maize, soybean, and other crops partially diverge as they approach the end of their growth stages. However, it remains uncertain whether these features alone can provide critical information for crop mapping during this period.

The temporal profile of each SAR feature suggests a similar linear relationship between the features and corresponding crop types. Consequently, assessing the nonlinear relationship between variables and targets, as well as the correlation between SAR features, becomes necessary. In addition, combining all of them increases feature dimensionality and computational cost. To address this, feature selection techniques were employed to filter out irrelevant features.

B. Feature Selection Outcomes

Boruta identifies optimal SAR features highly relevant to the prediction of target labels. As represented by gray circles (see Fig. 9), those SAR features with the highest importance scores (normalized to 1) were identified by the Boruta wrapper model at corresponding points in time. Based on gray circles, we, as the user, manually selected SAR features falling within

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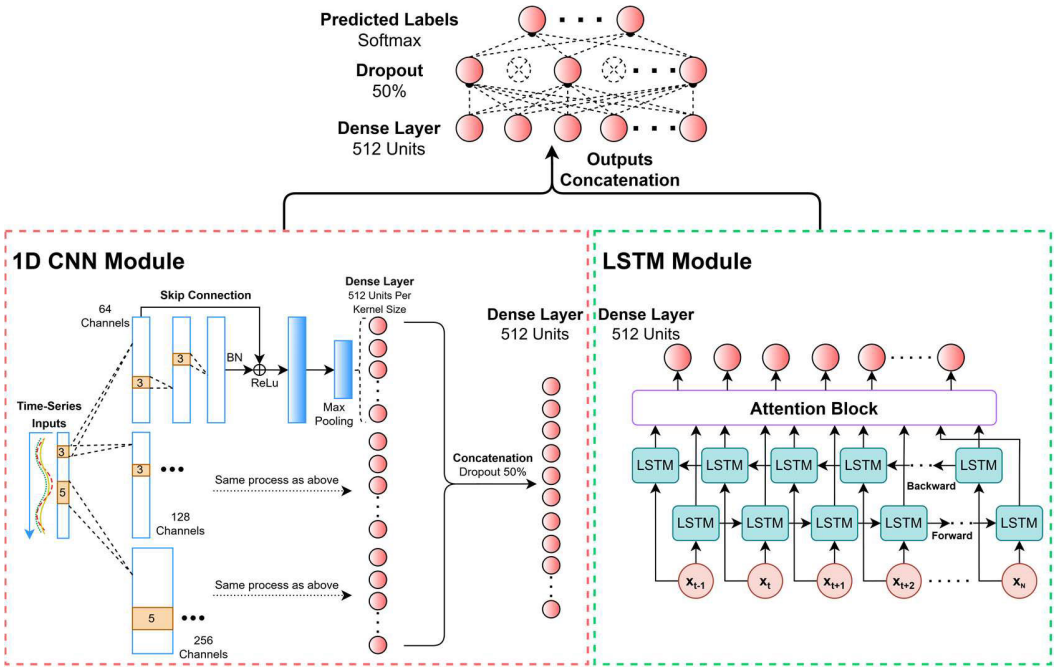


Fig. 7. Proposed joint learning network: Conv1D-LSTM.

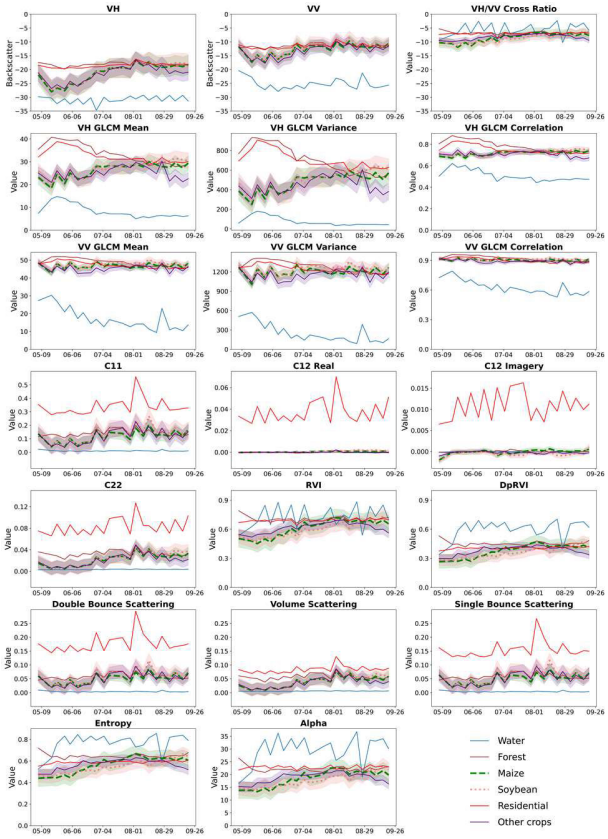


Fig. 8. Temporal profiles of Sentinel-1 input features. The buffer area in each subplot is one standard deviation from the mean for each crop type.



Fig. 9. Feature selection based on Boruta. Circles indicate the features with the highest importance scores, while full-time features are marked with stars. GLCM_c, GLCM_v, and GLCM_m indicate correlation, variance, and mean.

the entire crop growth cycle, which are crucial for in-season crop classification. These selected features are denoted by stars. Finally, the remaining features with complete temporal dimensions, such as DpRVI, GLCM, m-chi decomposition, and H/α dual-pol decomposition variables, were used to assess the multicollinearity among them.

The Spearman correlation coefficient (r_s) was calculated for each pair of selected features. This resulted in six pairwise combinations used to evaluate feature collinearity, as illustrated in Fig. 10. Each axis limit represents the total number of acquisitions for a pair of combined features, with emphasis placed on the second interval on the x-axis and the first interval on the y-axis in each subplot. These intervals contain subclasses of a full-time series SAR feature.

For instance, in the "DpRVI with m-chi" subplot, the total number of features across the growth stage is 88, displayed on both axes. The first interval represents 22 columns of sequential data for DpRVI, while the second interval indicates

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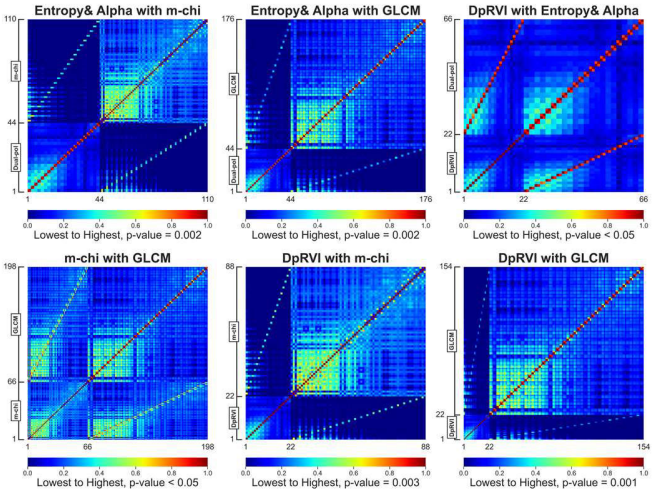


Fig. 10. Architecture of Conv1D and Conv1D-RF. Conv1D processes time series inputs and determines output classes with the softmax classifier. Conv1D-RF extracts inputs from the dense layer and predicts using the RF classifier.

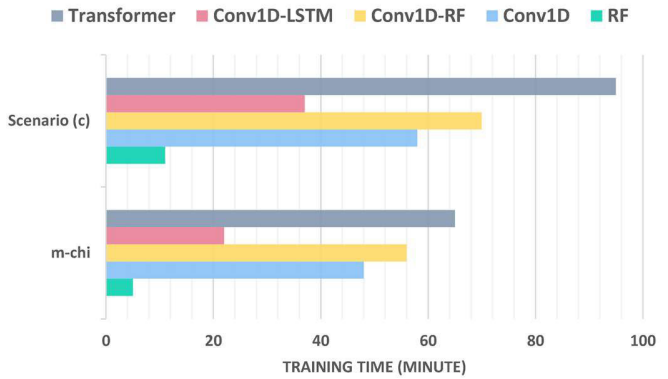


Fig. 11. Comparison of model training time for m-chi features and scenario (c) based on 10% of ground-truth training samples.

44 m-chi decomposition features. This arrangement forms rectangles on each axis. All subclasses of each feature are cross-calculated to determine the correlation strength, with r_s values derived from both features at the same time point represented by the off-diagonal lines of the rectangles. The intensity values between DpRVI and m-chi for the same date range fall between 0.4 and 0.6, indicating a moderate correlation.

The "DpRVI with Entropy & Alpha" pair exhibits stronger collinearity, with r_s values exceeding 0.8 and p-values less than 0.05, suggesting a high degree of correlation or linear dependency between the input variables. This implies that some features are redundant, providing overlapping information. Although deep learning models and RF are less sensitive to collinearity, three scenarios were employed to assess model accuracy and mitigate its impact: (a) "GLCM," "m-chi," and "Dual-pol"; (b) "GLCM," "m-chi," and "DpRVI"; and (c) a combination of all features. In addition, individual features were also evaluated for comparison.

C. Accuracy Assessment

The model performances are presented in Table II, revealing that deep learning models generally outperform RF in terms of OA and Kappa, as demonstrated by Conv1D, Conv1D-RF, Transformer, and the proposed Conv1D-LSTM. Specifically, Conv1D-LSTM achieves the highest OA (0.88) using only m-chi features, while Conv1D, Conv1D-RF, and Transformer yield similar results using the combined four features in scenario (c). Conv1D-LSTM also results in the second-best Kappa (0.83), close to the highest (0.84) in scenario (c).

This indicates that the diversity of input features may enhance accuracy for certain deep learning networks. However, models trained with optimal features can reduce computational costs for large-scale crop mapping. By selecting the most relevant and informative features, the complexity of the model is reduced, thus decreasing the time and resources required for training and prediction. This approach is particularly important for extensive datasets or when computational resources are limited. Consequently, scenario (c) exhibits redundancy in input predictors compared to using the m-chi variables alone.

Table III presents a comparison of model performance based on the $F1$ score for each class. Among the individual features, the m-chi feature demonstrates superior effectiveness for crop classification, as evidenced by higher $F1$ scores for models utilizing this feature compared to those using H/α, DpRVI, or GLCM. The proposed Conv1D-LSTM, when employing the m-chi features, exhibits the best performance, with the highest $F1$ scores for maize (0.85), soybean (0.84), other crops (0.82), and the highest average crop $F1$ score of 0.84.

In scenarios (a)–(c), the models generally achieve improved performance when combining multiple features, suggesting that incorporating a diverse set of input features can enhance overall $F1$ scores for certain deep learning networks. Scenario (c) is predominantly favored by Conv1D, Conv1D-RF, Transformer, and RF, exhibiting the highest $F1$ scores compared to other scenarios for identifying crops and land cover types.

The Conv1D-LSTM model using m-chi features highlights the importance of selecting optimal features to enhance computational efficiency in crop mapping and minimize redundancy in input predictors. The computational cost for model training with m-chi features is substantially lower than that of scenario (c), as illustrated in Fig. 11.

Therefore, all ground-truth samples of m-chi features were divided into training, validation, and testing sets with a 60%, 20%, and 20% ratio. The Conv1D-LSTM still yields the highest $F1$ scores for maize (87%), soybean (86%), and other crops (85%), outperforming other models, as shown in Fig. 12.

Overall, multistream deep learning architectures, such as the proposed Conv1D-LSTM and Conv1D-RF, exhibit better performance compared to other standalone architectures, including Conv1D and Transformer. All deep learning networks consistently outperform the traditional RF regarding the same metric for crop mapping. These results comprehensively explained the first research question.

Given that the optimal SAR feature is decided, the model generalisability was assessed across multiple small-scale regions within Bei'an. With 60% of all samples randomly selected from m-chi decomposition features for model training,

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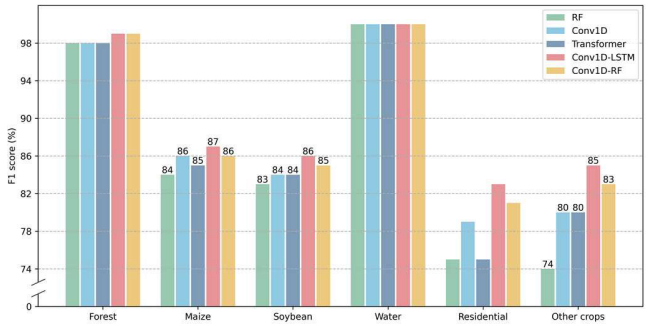


Fig. 12. Comparison of model performance based on m-chi features. Models were trained with 60% of all ground-truth samples and tested on the same testing set. Values are displayed for crops.

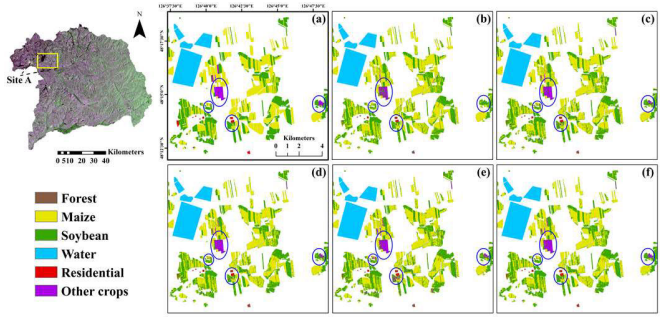


Fig. 13. Comparison of map predictions for Site A. (a) Reference labels. (b) RF. (c) Conv1D. (d) Conv1D-RF. (e) Transformer. (f) Conv1D-LSTM. Sentinel-1 false-color composite (blue: single-bounce scattering, green: double-bounce scattering, and red: volume scattering).

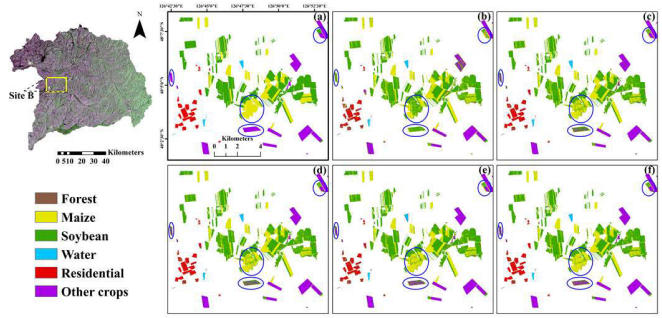


Fig. 14. Comparison of map predictions for Site B. (a) Reference labels. (b) RF. (c) Conv1D. (d) Conv1D-RF. (e) Transformer. (f) Conv1D-LSTM. Sentinel-1 false-color composite (blue: single-bounce scattering, green: double-bounce scattering, and red: volume scattering).

Conv1D, RF, Transformer, Conv1D-RF, and Conv1D-LSTM were employed for spatial map predictions and classification comparisons. The qualitative results for different geospatial locations are shown in Figs. 13–15.

Further post-classification procedures based on majority selection were conducted to reduce misclassified pixels in the prediction maps derived by Conv1D-LSTM (see Fig. 16), which demonstrates improved classification performance across all three sites compared to the proposed method alone. As a result, a county-scale classification map was produced by synergistically utilizing the Conv1D-LSTM pretrained based on 60% of all samples and the post-classification technique, as shown in Fig. 17. The improvements in classification performance can be observed in the confusion

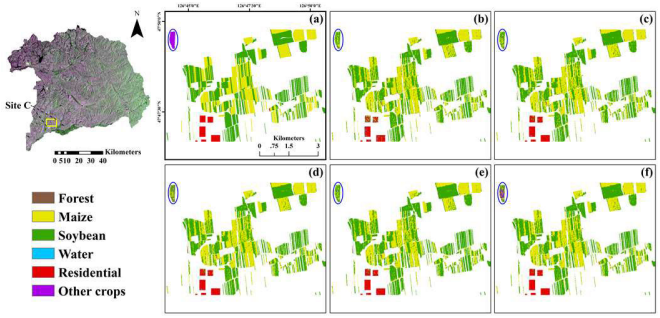


Fig. 15. Comparison of map predictions for Site C. (a) Reference labels. (b) RF. (c) Conv1D. (d) Conv1D-RF. (e) Transformer. (f) Conv1D-LSTM. Sentinel-1 false-color composite (blue: single-bounce scattering, green: double-bounce scattering, and red: volume scattering).

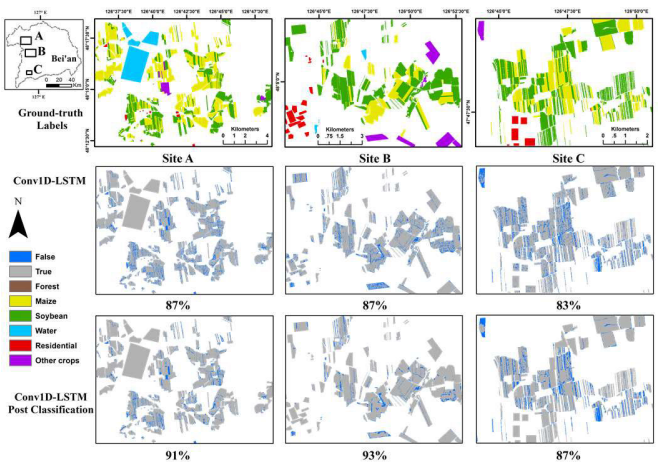


Fig. 16. Comparison of Conv1D-LSTM performance in three sample sites before and after post-classification. Percentages indicate the ratio of correctly classified pixels to ground-truth labels.

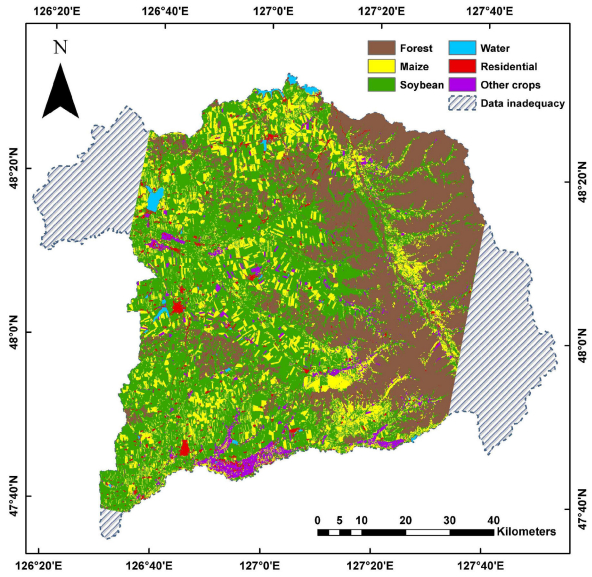


Fig. 17. Post-classification prediction map for Bei'an 2017. Data inadequacy refers to missing data resulting from incomplete coverage of the whole study area in Sentinel-1 SLC acquisitions across the growth season.

matrices when comparing all samples (see Fig. 18). Specifically, the recall values of maize, soybean, and other crops increased by 4%, 8%, and 7%, respectively, in the post-classification process, which suggests the effectiveness of

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TABLE II
ACCURACY ASSESSMENT FOR SELECTED FEATURES ON TESTING SETS. THE MODEL'S TRAINING, VALIDATION, AND TESTING WERE CONDUCTED USING 10% OF THE ENTIRE SAMPLE DATA. THE HIGHEST OA AND KAPPA VALUES ARE HIGHLIGHTED IN BOLD, AND THE SECOND-BEST VALUES ARE UNDERLINED

OA					
Features	Conv1D	Conv1D-RF	Conv1D-LSTM	RF	Transformer
H/α	0.74	0.74	0.74	0.72	0.74
DpRVI	0.68	0.72	0.72	0.71	0.73
GLCM	0.80	0.82	0.82	0.79	0.83
m-chi	0.85	0.85	**0.88**	0.84	0.86
Scenario (a)	0.86	0.86	0.85	0.83	<u>0.87</u>
Scenario (b)	0.86	0.86	0.84	0.83	0.86
Scenario (c)	**0.88**	**0.88**	0.85	0.84	**0.88**
Kappa					
Features	Conv1D	Conv1D-RF	Conv1D-LSTM	RF	Transformer
H/α	0.64	0.64	0.64	0.62	0.65
DpRVI	0.58	0.61	0.75	0.60	0.62
GLCM	0.73	0.75	0.75	0.71	0.77
m-chi	0.79	0.80	<u>0.83</u>	0.78	0.80
Scenario (a)	0.81	0.81	0.79	0.77	0.82
Scenario (b)	0.81	0.81	0.78	0.77	0.81
Scenario (c)	<u>0.83</u>	<u>0.83</u>	0.80	0.78	**0.84**

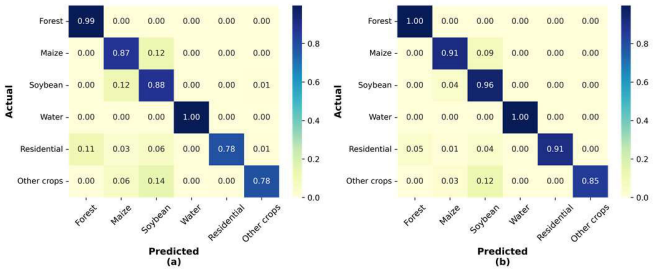


Fig. 18. Normalized confusion matrix for Conv1D-LSTM. (a) Preprocessed. (b) Postprocessed. Values are normalized as percentages for each class.

combining Conv1D-LSTM with post-classification techniques in improving the overall classification accuracy.

D. In-Season Crop Mapping

In-season crop mapping was assessed using monthly based temporal blocks corresponding to the phenology of major crops. This approach simulates a practical scenario, considering the increased availability of satellite acquisitions throughout the growing season. As shown in Fig. 12, the Conv1D-LSTM outperformed other Conv1D-based architectures, the machine learning model RF, and the Transformer model that relies on the self-attention mechanism. These models were compared with the Conv1D-LSTM in the assessment of in-season crop mapping (see Table IV).

The Conv1D-LSTM model surpassed the RF and Transformer models in terms of OA across all temporal windows. The OA for Conv1D-LSTM increased from 0.69 in May to 0.88 in September, highlighting the significance of the temporal dimension for sequential networks. For all models, the $F1$ scores for maize and soybean generally increased over the temporal windows, with the Conv1D-LSTM exhibiting the highest $F1$ scores for both crop types each month.

Improvements in maize were observed during the earlier growth stages. Conv1D-LSTM, RF, and Transformer improved by 8%, 9%, and 8% in June (tasseling), respectively, and by 9%, 6%, and 8% in July (ripening). By September (harvest), moderate improvements of around 3% were recorded for all models in August (end of flowering or silking) and September. Similarly, early growth stage improvements for soybean were 9%, 7%, and 7% in June (flowering) and 5%, 4%, and 6% in July (podding and seed filling) for each model. Values continued to improve by 5%, 4%, and 5% in August (matured), followed by slight improvements of 3%, 3%, and 2% in September (harvest).

The most notable increase in $F1$ scores for other crops was observed between May and June, with substantial improvements in June accounting for 42%, 33%, and 44% for Conv1D-LSTM, RF, and Transformer, respectively. August yielded the lowest improvement for other crops (Conv1D-LSTM: 5%, RF: −1%, and Transformer: 5%), but the transition from August to September improved models by 11%, 19%, and 7%.

E. Impact of Input Features on Model Performance

In addressing the second research question, this study found m-chi decomposition features to be efficient and effective in distinguishing crops, in agreement with previous findings (e.g., [33], [73], [74]). Sonobe [33] reported an $F1$ score of 0.84 for maize using RF, which was marginally lower than the scores of 0.85 (see Table III) and 0.87 (see Fig. 12) obtained in this study for maize before post-classification. However, they combined multiple predictors, ranging from 8 to 11, including the RVI, backscatter coefficients, and m-chi features, while this study relied solely on m-chi features. Although incorporating additional features could improve performance in scenario (c) in Table III, it could lead to extended model training time (see Fig. 11).

Using only m-chi features, this study's performance is comparable to that of [74], who produced recall values for maize (0.92) and soybean (0.79) using object-based RF, compared to maize (0.91) and soybean (0.96) derived in this study (see Fig. 18). In this study, covariance matrix parameters and backscatter coefficients were not included in the assessment of model accuracy according to Boruta (see Fig. 9); however, their importance still supports earlier findings in crop classification [26], [27], [66].

Previous studies have utilized fully polarimetric data based on coherency matrices (T_3) and quad-pol decomposition algorithms for multitemporal crop classification [30], [31]. They provide richer polarimetric information than C_2, increasing the likelihood of identifying complex land covers. Although exploiting quad-pol data could result in better performances than using dual-pol features for crop mapping, space-based quad-pol sensors are substantially limited by swath coverage [75], constraining their applicability for map prediction at the county level. In addition, obtaining time series quad-pol data might not be financially viable.

The in-season classification scheme operates model training based on incremental time series data along growth stages. Model performances in earlier growth stages were not statistically significant, suggesting that the SAR features available from maize with leaves and collars, as well as emerging

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TABLE III

MODEL PERFORMANCE BASED ON $F1$ SCORES. HIGHLIGHTED IN BOLD ARE THE HIGHEST $F1$ SCORES FOR CROP TYPES, AND THE SECOND-BEST SCORES ARE UNDERLINED. $\overline{\text{CROPS}}$ DENOTES THE AVERAGE $F1$ SCORES FOR ALL CROP CLASSES

Model	Features	Residential	Forest	Maize	Soybean	Water	Other crops	$\overline{\text{Crops}}$
Conv1D [41]	H/α	0.74	0.84	0.71	0.71	0.98	0.45	0.62
	DpRVI	0.29	0.80	0.59	0.69	0.92	0.35	0.54
	GLCM	0.58	0.97	0.73	0.77	1.00	0.67	0.72
	m-chi	0.70	0.98	0.82	0.80	0.99	0.70	0.77
	Scenario (a)	0.77	0.98	0.84	0.82	1.00	0.75	0.80
	Scenario (b)	0.75	0.98	0.82	0.82	1.00	0.75	0.80
	Scenario (c)	0.81	0.99	**0.85**	**0.84**	1.00	0.79	0.83
Conv1D-RF [54]	H/α	0.61	0.84	0.71	0.72	0.98	0.48	0.64
	DpRVI	0.37	0.82	0.70	0.70	0.97	0.43	0.61
	GLCM	0.64	0.97	0.78	0.77	1.00	0.71	0.75
	m-chi	0.75	0.98	0.82	0.81	1.00	0.74	0.79
	Scenario (a)	0.76	0.98	0.83	0.82	1.00	0.75	0.80
	Scenario (b)	0.75	0.98	0.83	0.82	1.00	0.75	0.80
	Scenario (c)	0.80	0.99	**0.85**	**0.84**	1.00	0.79	0.83
RF	H/α	0.26	0.83	0.70	0.70	0.96	0.37	0.59
	DpRVI	0.15	0.81	0.69	0.69	0.96	0.37	0.58
	GLCM	0.34	0.96	0.75	0.74	1.00	0.58	0.69
	m-chi	0.62	0.97	0.81	0.80	1.00	0.64	0.75
	Scenario (a)	0.56	0.97	0.81	0.80	1.00	0.59	0.73
	Scenario (b)	0.54	0.97	0.81	0.79	1.00	0.60	0.73
	Scenario (c)	0.66	0.97	0.82	0.81	1.00	0.66	0.76
Conv1D-LSTM	H/α	0.65	0.84	0.71	0.72	0.98	0.44	0.62
	DpRVI	0.36	0.82	0.70	0.70	0.97	0.43	0.61
	GLCM	0.58	0.97	0.78	0.76	1.00	0.69	0.74
	m-chi	0.79	0.98	**0.85**	**0.84**	1.00	**0.82**	**0.84**
	Scenario (a)	0.73	0.98	0.83	0.81	0.99	0.63	0.76
	Scenario (b)	0.63	0.97	0.81	0.81	1.00	0.67	0.76
	Scenario (c)	0.74	0.98	0.83	0.81	1.00	0.70	0.78
Transformer [45]	H/α	0.65	0.84	0.72	0.72	0.98	0.48	0.64
	DpRVI	0.45	0.82	0.70	0.70	0.97	0.45	0.62
	GLCM	0.66	0.98	0.80	0.78	1.00	0.73	0.77
	m-chi	0.67	0.97	0.83	0.82	1.00	0.72	0.79
	Scenario (a)	0.77	0.98	0.84	0.83	1.00	0.76	0.81
	Scenario (b)	0.75	0.98	0.83	0.82	1.00	0.76	0.80
	Scenario (c)	0.75	0.99	**0.85**	**0.84**	1.00	0.80	0.83

TABLE IV

IN-SEASON CROP MAPPING PERFORMANCE. MODEL TRAINING, VALIDATION, AND TESTING WERE BASED ON 10% OF ALL SAMPLE DATA OF M-CHI DECOMPOSITION FEATURES. HIGHLIGHTED IN BOLD ARE THE BEST RESULTS

Temporal windows			May	June	July	August	September
Conv1D-LSTM		OA	0.69	0.76	0.82	0.85	**0.88**
	F1	Maize	0.61	0.69	0.78	0.82	**0.85**
		Soybean	0.62	0.71	0.76	0.81	**0.84**
		Other crops	0.12	0.54	0.66	0.71	**0.82**
RF		OA	0.68	0.75	0.79	0.82	0.84
	F1	Maize	0.60	0.69	0.75	0.79	0.82
		Soybean	0.62	0.69	0.73	0.77	0.80
		Other crops	0.09	0.42	0.46	0.45	0.64
Transformer		OA	0.68	0.74	0.80	0.84	0.86
	F1	Maize	0.59	0.67	0.75	0.81	0.83
		Soybean	0.62	0.69	0.75	0.80	0.82
		Other crops	0.03	0.47	0.60	0.65	0.72

soybeans, were insufficient for effective model extraction. The poor performance of other crops in the early season can be attributed to the impact of bare soil. However, performance gradually improves as acquisitions corresponding to phenological stages are continuously accumulated. This supports the hypothesis that consecutive temporal windows in line with complete growth stages are inherently beneficial for enhancing crop mapping. This finding appears to be consistent with previous studies [46], [51], which demonstrated that each growth stage of observations in a complete time series enriches data structures during model training. This corresponds to the third research question. Although all models benefited from sequentially accumulated SAR acquisitions, Conv1D-LSTM consistently outperformed RF and Transformer on a monthly basis in terms of recognizing all crop types.

F. Interpretation of Learning Behavior of Conv1D-LSTM

The interpretability of deep learning models remains a challenge in remote sensing studies. In the end-to-end neural networks, neuron weights are iteratively updated and optimized through backpropagation to minimize errors between inputs and targets. The full set of mathematical operations is concealed during model training, resulting in complexities and difficulties in outcome analysis, commonly known as the "closed box." Although many studies have thoroughly explained Conv1D-based architectures and attention-based LSTM in terms of their learning behaviors with time series

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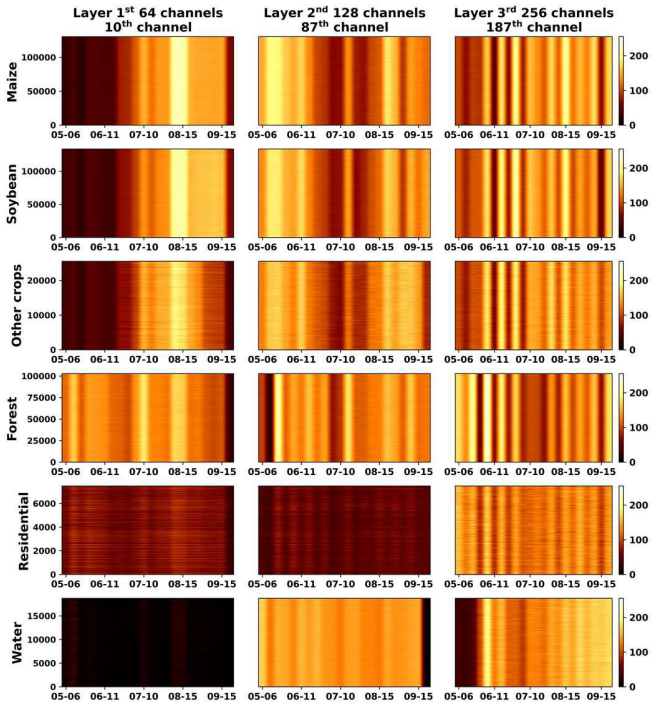


Fig. 19. Visualization of feature maps from three-level Conv1D layers in Conv1D-LSTM. The output feature maps were extracted from the model pretrained with m-chi features. The y-axis represents the sample count for each class in the testing set, while the x-axis displays Sentinel-1 acquisitions at monthly intervals. The weight values' range is normalized from 0 to 255 to facilitate visualizing intensity for weight distributions within each channel.

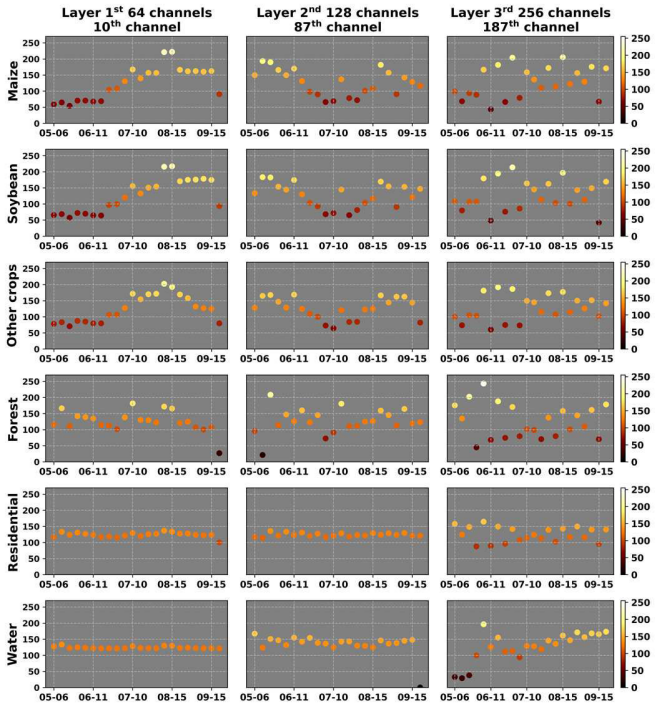


Fig. 20. Visualization of feature maps based on average weight distribution across multiscale Conv1D layers. The range of weight values is normalized from 0 to 255 on the y-axis.

multispectral data for crop classification [41], [45], [52], [54], the proposed Conv1D-LSTM, due to its intricate design, necessitates interpretation of the learned hidden features in the 1D-CNN and LSTM module to respond the fourth research question.

The visualization of the learning behavior of the 1D-CNN end of Conv1D-LSTM is intuitively assessed in Fig. 19. Weight distribution along time steps for each class is visualized over multilevel Conv1D layers. Selected convolutional layers include the first layer with 64 channels, the second layer with 128 channels, and the third layer with 256 channels, allowing for interpretation of the model on multiscale feature learning. For crops, weights are mostly distributed around "August 15" in the first layer, suggesting that important acquisitions could be roughly identified by the model's shallow layer. As feedforward propagations advance through the following layers, weight distributions become more localized, and important acquisitions are identified at specific time steps. For instance, more acquisitions are intensively weighted around June (tasselling for maize and flowering for soybean) and July (maturing) for crops.

Another visualization of weight distribution is based on the average weight on each time step (see Fig. 20). Weight intensities appear to fluctuate diffusely throughout the entire growth stage as the layer goes deep, with important acquisitions identifiable at the third Conv1D layer for each class. However, regarding the trends for crops, their similar weight profiles may still contribute to high rates of model misclassification. This can be modulated with the LSTM module of Conv1D-LSTM.

The attention-based LSTM in Conv1D-LSTM generates attention weights that can be observed in Fig. 21. A distinct difference in weight profiles for maize and soybean is evident, potentially enhancing the separability between these two crops. This emphasizes the significance of m-chi features in distinguishing crops based on the model's decisions. The attention weights for maize and other crops start to increase rapidly from "July 10" as they enter the maturation stage, implying that cumulative information becomes progressively more useful for crop classification from this point onward. For soybean, the same turning point is observed around "August 15" during its maturity stage. These results align with the findings of [52], who also identified rapidly increasing attention weights for maize at the silking stage on "July 15" (following the tasselling stage) and for soybean on "August 19." Moreover, it is plausible that the observed increase in the $F1$ score for other crops in September (see Table IV) may be connected to the rising attention weights within the same temporal window since the weight intensities are not significant in the Conv1D layers around "September 15." This observation could help elucidate the attention mechanism's contribution to classifying the minority class.

The model performance is not only reflected in the model accuracy assessment but can also be visualized through learned hidden features. The t-distributed stochastic neighbor embedding (t-SNE) visualization technique [76] allows for dimensionality reduction on high-dimensional features.

In this study, the high-level hidden features are 512-D and exhibit complex patterns for multitemporal information. Analyzing such a data structure intuitively is challenging, but t-SNE enables the nonlinear projection of hidden features onto

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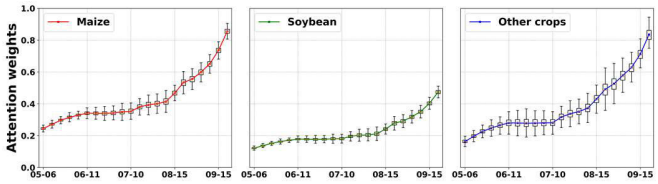


Fig. 21. Crop attention weight profiles derived from the LSTM module in Conv1D-LSTM. The output attention weights were obtained using m-chi features. Values on the y-axis are scaled to a range from 0 to 1 for improved visualization.

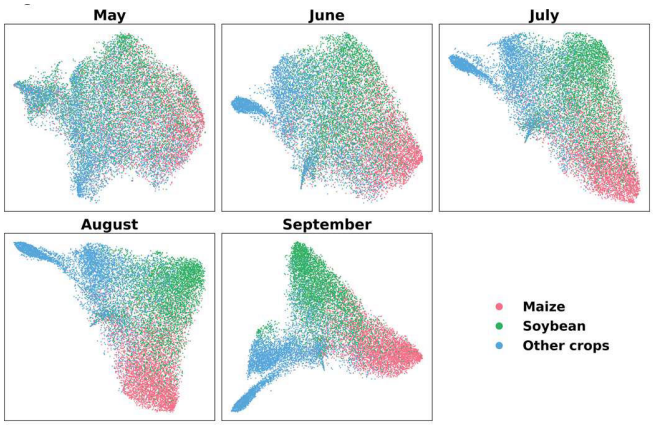


Fig. 22. Visual comparison based on t-SNE along monthly blocks for learned hidden features, wherein 5000 randomly selected neural samples from each crop type were extracted from the MLP unit of the Conv1D-LSTM model to enhance visualization. Each point represents a single neural sample corresponding to a specific crop category.

a 2-D plane for visual comparison of crops' separability along monthly blocks (see Fig. 22).

In the early stage in May (emerging), the learned neural samples of all crops are closely situated. As the growth stages progress, these samples become increasingly segregated and grouped according to crop types. From June (tasselling and flowering) onward, crop separability expands toward the harvest stage in September, demonstrating that the accumulation of temporal information contributes to improving crop classification performance. These findings align with the in-season classification results presented in Table IV. Although t-SNE is not always a conclusive indicator for supporting quantitative analysis of learned features, it provides an alternative way to intuitively examine the outcomes of the hidden learning patterns from deep learning models.

G. Potential Extension of This Study

In this study, the dual-polarization system of Sentinel-1 SLC has demonstrated its potential to generate pseudo-compact PolSAR parameters, which, to a certain extent, can improve the discrimination for crops with phenological similarities. This is achieved with the aid of the proposed joint ensemble learning Conv1D-LSTM for enhanced classification performance. This suggests that compact polarimetry holds promise for improving land cover classification. Moreover, there is a need for an expanded time series to correlate with more detailed phenological phases of crops. This could be accomplished by incorporating optical acquisitions that align with key SAR acquisitions identified in this study, which relates to the synergistic use of SAR and multispectral features for improved crop mapping.

Further optimization of the model architecture and classification workflows can enhance multitemporal crop mapping. For instance, Conv1D-LSTM focuses merely on pixel-based feature extraction from time series data, disregarding spatial relationships between features. However, 3D-CNN architectures, which have shown promise in crop classification using optical data, take both temporal and spatial dimensions into account. Exploring this approach for crop mapping that combines SAR and optical data could lead to further improvements [77], [78].

Another promising avenue for model development is the integration of data-driven models and physical models for agricultural applications using remote sensing data. Physical models offer strong interpretability and performance but often contain redundant parameters and lack efficiency. In contrast, deep learning models suffer from limited interpretability. Combining both model mechanisms could create a mutually complementary effect, addressing each other's limitations. Existing studies have applied this combination by replacing a submodule of a physical model with RF [79] or by modulating the loss function in the deep learning model using physical mechanisms [80]. It would be promising to examine how these approaches could improve crop mapping performance.

In addition to model design, it is also essential to evaluate the spatiotemporal transferability of the models in crop mapping, considering interregional and interannual variability. Several factors can influence the transferability, such as differences in crop growth conditions, cropping practices (e.g., crop rotation and intercropping), spatial heterogeneity, and the representativeness of the training data. Mismatches in time series data between training and testing regions due to these factors can lead to poor classification performance. To address this, transfer learning techniques like fine-tuning with new data can help adapt pretrained models to the specific agricultural conditions and cropping patterns of the test regions. Furthermore, incorporating additional dynamic features that capture the temporal and spatial variability of croplands, such as environmental parameters and meteorological variables [4], [81], may further improve the models' predictive performance and transferability across different geographical areas and temporal contexts.

V. Conclusion

This study demonstrated the effectiveness of using joint ensemble learning of two temporal models to extract multitemporal features from dual-pol SAR data for predicting county-level crop categories in Bei'an for 2017. The proposed Conv1D-LSTM model outperformed Conv1D, Conv1D-RF, Transformer, and baseline RF models. With optimally selected SAR features (m-chi decomposition), Conv1D-LSTM achieved the highest $F1$ scores for maize (87%), soybean (86%), and other crops (85%). The model effectively handled imbalanced data and differentiated between summer crops with similar phenology. The in-season classification results emphasized the importance of multitemporal information for

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crop classification, with Conv1D-LSTM capturing temporal dependencies across growth stages, leading to superior monthly performance. Post-classification further improved performance. Visualizations of weight distributions highlighted critical phenological stages, and hidden feature analysis showed the learning impact of Conv1D-LSTM over time. This study concludes that joint learning of Conv1D and attention-based LSTM can produce accurate CDLs at a large scale. Future work will focus on designing a spatiotemporal learning approach to transfer reliable pixels and pretrained models from Bei'an to other regions.

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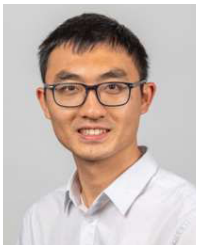
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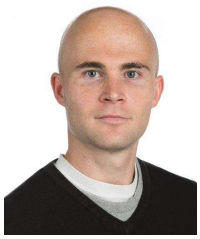
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